Practice Numerical Problems on Diffraction

1. A plane transmission grating produces a second-order principal maximum for light of wavelength 600 nm at a diffraction angle of 30°. Calculate the grating element of the grating.
2. Calculate the highest order spectrum that can be obtained using monochromatic light of wavelength 7000 Å with a grating having 8000 lines/cm.
3. A diffraction grating at normal incidence gives a line of wavelength 6000 Å in the n^{th}order, superimposed on a line of wavelength 4800 Å in the next higher order. If the diffraction angle is 30°, calculate the number of lines per cm
4. The grating is 6000 rulings/cm and is 3 cm wide. Calculate the smallest wavelength interval that can be resolved in the third order at a wavelength of 600 nm.
5. Find the maximum resolving power of a grating of width 2 cm having 5000 lines/cm, when illuminated with light of wavelength 6000 Å